# Distributed Linear Quadratic Optimal Control: Think Locally and Act Globally

Junjie Jiao, Harry L. Trentelman and M. Kanat Camlibel

Bernoulli Institute for Mathematics,
Computer Science and Artificial Intelligence

University of Groningen

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# Think Locally and Act Globally

What do I mean by

'Think Locally and Act Globally'?

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Explain later.

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# Distributed LQ Control Problem

- Multi-agent system:

$$\dot{x}_i(t) = u_i(t), \quad x_i(0) = x_{i0}, \quad i = 1, 2, \ldots, N$$

# Distributed LQ Control Problem

Let $x = (x_1, x_2, \ldots, x_N)^\top$ and $u = (u_1, u_2, \ldots, u_N)^\top$.

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- Multi-agent system:

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- Connected undirected graph
- General cost functional:

$$J(x_0, u) = \int_0^\infty x^\top(t) LWLx(t) + u^\top(t) Ru(t)\ dt, \quad W \geq 0,\ R > 0$$

Note: This involves a weighted norm $||Lx(t)||_W^2$, while $Lx(t) \to 0$ as $t \to \infty$ means consensus.

# Example 1

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$$J(x_{i0}, u_i) = \int_0^\infty \frac{1}{2} \sum_{i=1}^{N} \sum_{j \in \mathcal{N}_i} q\Big(x_i(t) - x_j(t)\Big)^2 + \sum_{i=1}^{N} r u_i^2(t)\ dt$$

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$$J(x_0, u) = \int_0^\infty x^\top(t) L q L^\dagger L x(t) + r u^\top(t) u(t)\ dt$$

where $W = qL^\dagger$ and $R = rI_N$.

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# Distributed LQ Control Problem

- Multi-agent system:

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- Undirected connected graph
- General cost functional:

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- Control protocol:

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After substituting (2) into the cost functional (1), we obtain $J(g)$.
The problem is to minimize $J(g)$ over all $g > 0$.

# Solution to Distributed LQ Control Problem

## Proposition 1

*It holds that*

$$J(g) = \frac{1}{g} x_0^\top X_0 x_0 + g x_0^\top Y_0 x_0.$$

*where* $X_0 \geq 0$ *and* $Y_0 \geq 0$ *are special solution of*

$$-LX - XL + LWL = 0, \quad -LY - YL + LRL = 0.$$

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$$-LX - XL + LWL = 0, \quad -LY - YL + LRL = 0.$$

The cost depends on the initial states of all agents and the Laplacian matrix, which is global information.

The solution $g^*$ depends almost everything of the entire network.

# Drawback of Global Cost Functional

The computation of the optimal gain needs

1. global information:
   - the initial states of all agents, $x_0 = (x_{10}, x_{20}, \ldots, x_{N0})^\top$
   - the Laplacian matrix $L$

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1. global information:
   - the initial states of all agents, $x_0 = (x_{10}, x_{20}, \ldots, x_{N0})^\top$
   - the Laplacian matrix $L$
2. highly impractical and computation demanding:
   if some agents leave or new agents join the group, the computation of the optimal gain has to be repeated all over again.

Therefore, it is in practical impossible to compute the optimal gain!

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- Think Locally: compute locally.
  That is, each agent computes its own local control gain by using local information only.
- Act Globally: every agent computes its own local control law, which results in a distributed protocol for the overall network.

# Example 2 - Revisit

- Multi-agent system:

$$\dot{x}_i(t) = u_i(t), \quad x_i(0) = x_{i0}, \quad i = 1, 2, \ldots, N$$

- Global cost functional:

$$J(x_{i0}, u_i) = \sum_{i=1}^{N} \int_0^\infty q\Big(x_i(t) - a_i(t)\Big)^2 + r u_i^2(t)\ dt$$

with

$$a_i(t) = \frac{1}{d_i + 1}\big(x_i(t) + \sum_{j \in \mathcal{N}_i} x_j(t)\big).$$

# Local Cost Functionals

- Multi-agent system:

$$\dot{x}_i(t) = u_i(t), \quad x_i(0) = x_{i0}, \quad i = 1, 2, \ldots, N$$

- Local cost functional:

$$J_i(x_{i0}, u_i) = \int_0^\infty q\,(x_i(t) - a_i(t))^2 + r u_i^2(t)\ dt, \qquad i = 1, \ldots, 6$$

with

$$a_i(t) = \frac{1}{d_i + 1}\big(x_i(t) + \sum_{j \in \mathcal{N}_i} x_j(t)\big).$$

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1. At time $t = 0$, agent $i$ receives $x_j(0)$ of its neighbor agents and compute the average

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2. Agent $i$ minimizes the cost functional

$$J_{i,0}(u_{i,0}) = \int_0^\infty e^{-2\alpha t}\left(q\left(x_i(t) - a_i(0)\right)^2 + r u_{i,0}^2(t)\right) dt.$$

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The above procedure is repeated for $[kT, (k+1)T,\ k = 0, 1, 2, \ldots$.

# Decentralized Computation

If all agents compute their own local control laws, this gives us a distributed protocol for the entire multi-agent system.

**Theorem 1**

*The distributed protocol achieves consensus for the controlled network.*

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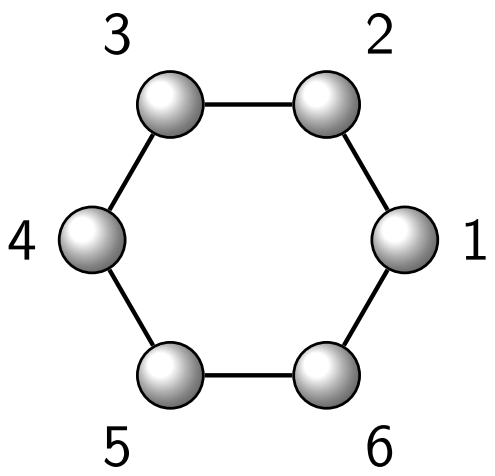


- Agent dynamics:

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- Local cost functionals:

$$J_i(u_i) = \int_0^\infty q\left(x_i(t) - a_i(kT)\right)^2 + r u_i^2(t) dt, \quad i = 1, \ldots, 6$$

with $\quad a_i(kT) = \dfrac{1}{d_i + 1}\big(x_i(kT) + \sum_{j \in \mathcal{N}_i} x_j(kT)\big), \quad k = 0, 1, \ldots.$

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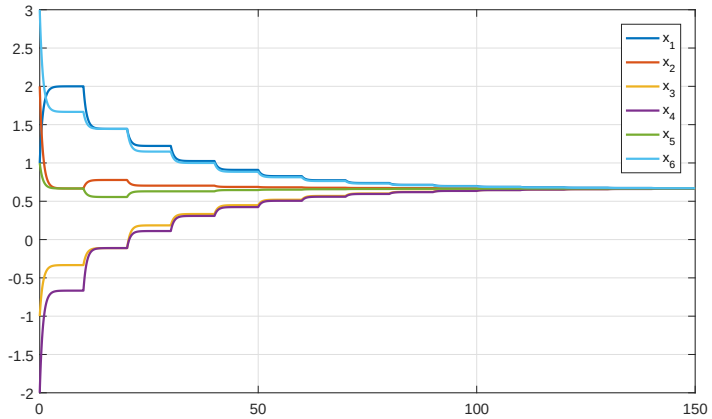


Figure: Plot of the states of the controlled network with $T = 10$

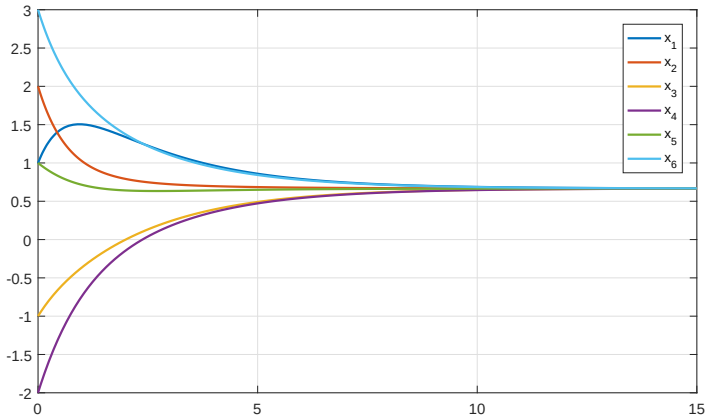


Figure: Plot of the states of the controlled network with $T = 0.1$

For the sampling period $T = 0.1$, comparing to $T = 10$,

- it takes a larger number of required iterations to reach a small state disagreement,
- but still leading to a smaller total time.

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1. Distributed LQ control problem with global cost functional:
   - highly depends on global information,
   - computationally expensive, in practical impossible.
2. Alternative approach to compute the local control gains:
   - the computation is decentralized, scalable in design
   - achieves consensus.

Thank you!